FOR FLAT FINITELY PRESENTED MAPS, LCI, SYNTOMIC, AND THE NL TOR-AMPLITUDE [-1,0] CONDITIONS ARE ALL EQUIVALENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18500-18510 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18512-18522, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The syntomic-side equivalence used throughout the chapter's lci/syntomic formalism (pairs with collected lci-properties, regular-immersion-lci cards).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas affine-lci and NL-affine; Derived Categories of Schemes, lemmas affine-compare-bounded, tor-dimension-affine, perfect-affine; Divided Power Algebra, lemma flat-fp-NL-lci.

Source reference. Original lemma 0FJV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18522–18522. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D0I: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3736–3745. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07DB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18178–18184. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06Z0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18522–18522. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08EB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18522–18522. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 08E9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18522–18522. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a flat morphism of finite presentation. The following are equivalent*

(1) *f is a local complete intersection morphism,*
(2) *f is syntomic,*
(3) *$NL_{X/Y}$ has tor-amplitude in $[-1, 0]$, and*
(4) *$NL_{X/Y}$ is perfect with tor-amplitude in $[-1, 0]$.*

Proof. Translated into algebra this is Divided Power Algebra, Lemma 0FJV. To do the translation use Lemmas 07DB and 0D0I as well as Derived Categories of Schemes, Lemmas 06Z0, 08E9 and 08EB. □